From Frequency Response to Bode

EGR 345

GVSU

Outline

- quiz solution
- review
- from frequency response to Bode plots
  - from C_1 and C_2 to M and ϕ
  - what if we do lots of frequencies? (LA 3)
  - are we ready for `semilogx` and dB magnitude?

Review 1

- if we give a sine wave input to a stable system, what happens?

Review 1B

- $u(t) = \sin(\omega t)$
- $G(s)$ is stable
- $y_{ss}(t) = ?$

Review 2

- if the input frequency changes, how does y_{ss} change?
  - what did you observe in lab?
    - simulation?
    - experiment?

Review 3

- how did we find C_1 and C_2?

How do we find magnitude and phase?

- M is magnitude and ϕ is phase

$$y_{ss}(t) = C_1 \sin(\omega t) + C_2 \cos(\omega t) = M \sin(\omega t + \phi)$$

$$\sin(\alpha \pm \beta) = \sin\alpha \cos\beta \pm \cos\alpha \sin\beta$$

What is a Bode plot?

LA 3

- let’s do this